

AIML 102

B. Tech. (IInd SEMESTER) EXAMINATION, 2025-26

(CSE-AIML)

Conversational AI

Time : Three Hours]

[Maximum Marks : 75

Note: There are **three** sections (A, B and C) and candidate has to attempt questions from all sections. Marks are indicated against each section.

Section-A

1. Attempt **all** parts of the following : 5×3=15
- (a) Explain the Conversational Design Process and importance of writing the script.
 - (b) Define Intents, Entities and training phrases in interaction models.
 - (c) What is the difference between Actions Parameters and slot filling ?
 - (d) What is web hook ? Explain its role in conversational AI.
 - (e) Explain Re-training, Validation and Testing after deployment.

Section-B

Note: Attempt **all** questions :

5×4=20

2. (a) Explain use cases of chatbot / virtual assistant and explain Conversational Design in details.

Or

- (b) Describe the steps to set up Dialog Flow for a chatbot project.

3. (a) Explain Agents, rich responses, small talk and salutations.

Or

- (b) How do you configure and test Intents on Google Assistant ?

4. (a) Differentiate between Linear and Non-linear Dialogue with examples and diagrams.

Or

- (b) Explain Actions and Parameters and their use in extended lead Generation.

5. (a) Explain basic setup of webhook code with parameter extraction.

Or

- (b) Explain Fulfilment using Cloud Function with a suitable example.

Section-C

Note: Attempt any **two** questions :

2×20=40

6. (a) Draw and explain a complete conversational flow for a customer-support chatbot, including script writing and integration with Dialog flow.
- (b) Explain the role of Agents, Intents, Entities and training phrases while building interactions models for the University admission chatbot.
7. (a) A healthcare clinic wants a chatbot for appointment booking and symptom checking. Explain linear Vs Non-linear Dialogue with suitable flow diagrams for this real-life healthcare scenario.
- (b) How will slot filling, context and Actions & Parameters help in making the healthcare chatbot conversation more user-friendly ? Illustrate with example.

8.
 - (a) Explain the full deployment process of the healthcare chatbot on Heroku as well as on Alexa. Compare both platforms for this practical healthcare use cases.
 - (b) After successful deployment, explain why Re-training, Validation and Testing are essential. Describe the complete testing and validation steps you would perform before making the chatbot live for patients.
9.
 - (a) Explain Webhook in detail. Write the basic webhook code for extracting parameters and structuring responses.
 - (b) Differentiate between webhook - based fulfilment and Cloud Function fulfilment. When would you prefer one over the other ?